

Using immersive video to recall significant musical experiences in elderly population with intellectual disability

Pablo Pérez*
XR Lab, Nokia

Marta Orduna
XR Lab, Nokia

María Nava-Ruiz
Fundación Juan XXIII

Javier Martín-Boix
Fundación Juan XXIII



Figure 1: Left: recording of one of the music sessions. Center: view of the session from the VR headset. Right: One of the participants during the sessions.

ABSTRACT

Comodia Elderly project studies the use of immersive video in elderly population with intellectual disability. Participants in the project took part in live concerts, which were recorded with a 180-degree stereoscopic camera. Recording were visualized in successive sessions of virtual reality video. Preliminary results show a high level of spatial and social presence, assessed both from adapted questionnaires and for external observation of observable signs.

Index Terms: Social and professional topics—People with disabilities—; Information systems—Multimedia information systems—

1 INTRODUCTION

Median life expectancy of people with intellectual disabilities, such as those with Down Syndrome, has significantly increased from 4 years in the 1950s to 58 years by 2010 [1]. This not only extends their lifespan but also enhances their quality of life with more diverse experiences. Virtual reality can provide these individuals with unique, otherwise inaccessible experiences, broadening their emotional and experiential world.

In a previous experience (project Amatista), we showed that immersive music therapy based on 360-degree video can be beneficial for elderly population with mild to moderate cognitive disease [4]. In particular, we observed that the reported levels of spatial and social presence in such population are similar to the ones observed in university students, and they are not correlated with the level of cognitive degradation observed in such patients.

Following that experience, we launched Comodia Elderly project to validate if the results could be translated to ageing population with intellectual disability. Specifically, population from the daycare center of Fundación Juan XXIII in Madrid, Spain. The project aims to study the impact of immersive technologies and telepresence in

elderly individuals with intellectual disabilities, defined as those over 35 years old with Down Syndrome and over 45 without it.

Comodia Elderly project objectives include enhancing life satisfaction, fostering positive emotions, reducing negative feelings, aiding memory recall of significant events, and preserving cognitive functions. The study anticipates several benefits for participants, such as experiencing musical events, accessing new technologies, enjoying pleasant emotions through music, visiting desired locations virtually, connecting with important people, improving overall life satisfaction with novel activities, and triggering memories through these experiences.

2 PROJECT ACTIONS

2.1 Selection and evaluation of participants

The selection of participants was based on a series of meetings held by the professional team of the Fundación Juan XXIII. This selection took into consideration their extensive knowledge of the interests and needs of the users who met the aging criteria described earlier. An informal interview with the potential participants was also conducted to explain the use cases designed for them and to assess their interest and motivation to participate. 12 participants were selected between 39 and 75 years old ($M = 57$, $SD = 9.9$), 3 female and 9 male, 3 with Down Syndrome.

Before the start of the experiences, an initial evaluation of all participants was carried out to establish a baseline. This evaluation included questionnaires on life satisfaction, affect, cognitive decline, and autobiographical memory. Participants exhibited a high degree of life satisfaction and a tendency towards a positive affective state, possibly due to the care received at the daycare center, which aims to achieve a high degree of satisfaction in all quality of life dimensions. However, significant differences were found in the number of memories evoked related to autobiographical memory. It is expected that after the completion of the project, there would be an increase in the number of evoked memories, as well as improvements in the scores of the other scales.

2.2 Recording of musical events

There were 10 recordings of musical group performances (Fig. 1 left), which were of interest to the participants and previously identified in a joint exploratory session. Each recording consisted of

*e-mails: pablo.perez@nokia.com, marta.orduna@nokia.com, mnava@fundacionjuan23.org, jmartin@fundacionjuan23.org

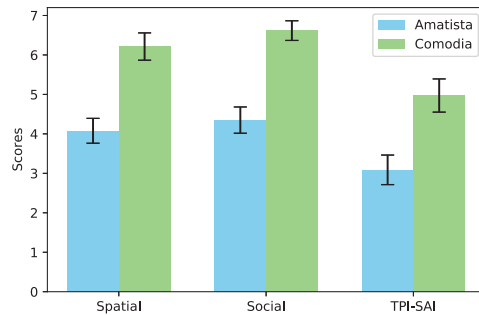


Figure 2: Mean and 95% CI of spatial, social, and social-active presence scores of elderly with intellectual disability (Comodia Elderly), compared with elderly with neurodegenerative diseases (Amatista) [4].

6 songs, each lasting approximately 2-3 minutes, from different musical groups. All the elderly at the daycare center, including the 12 participants, attended the live concerts. The artists gave up their rights for the use of this research within the framework of this project.

Concerts were recorded in 180° stereoscopic video using Vuze XR, located in the scenario. Equirectangular panorama, 4K resolution was used. Audio was captured from the microphone mix of the recording. Produced contents were compressed in H.264 and streamed in HLS. Even though the full bitrate ladder was generated, it was ensured that only the highest quality (3840x2160, 20 Mbps) was used in the tests.

2.3 Immersive Sessions

Weekly immersive video sessions were conducted in groups of 4 people (12 participants in total, divided into 3 groups). Each session featured 4 songs with a total duration of about 12 minutes. Each participant had a total of 4 sessions and an overall immersive experience of around 45 minutes. Each one had their own individual equipment: Meta Quest 2 and headphones. The application from [4], with some modifications, was used to display the video. A control web application was in charge of executing each session playlist, by sending a message to each of the VR Headsets whenever a new musical piece was started.

In each session, the sense of presence was assessed using the protocol defined in [4]. The therapist recorded the observable signs that indicated a sense of immersion in the video, using the Active Social Presence subscale of the Temple Presence Inventory [2]. Additionally, Spatial and Social Presence were evaluated after each session using direct questions [3]. The questions were adapted to the population with the help of the therapists. Scoring of each item was reduced to 3 levels (*a lot, some, a little*), represented by pictograms (smileys). It is familiar to users and allows some of them score touching the screen, minimizing professional intervention.

3 RESULTS

Fig. 2 shows the presence measures compared with [4] (age $M = 87.8$, $SD = 5.5$). Fig. 3 shows the evolution of results in the first four sessions. All scores are scaled to 7 levels for comparison with [4].

There is a discernible sense of presence in the study. The presence reported is significantly higher compared to individuals in [4]. Both the experience and the measurement protocol are similar: we can conclude that both populations show significant difference in reported presence. Some factors that can explain this difference are: i) Comodia population normally report over-optimistically when asked to evaluate a experience; ii) they are much more expressive and extrovert; iii) they had already taken part in live concerts of the same artists, so they were recalling a specific significant experience.

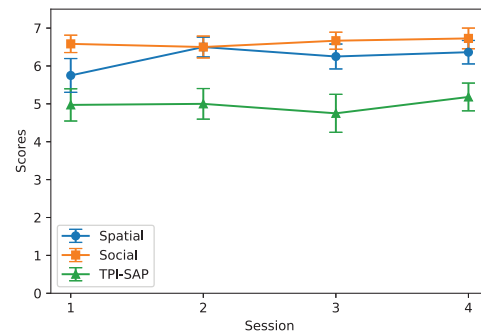


Figure 3: Mean and standard error of the three presence measures across the sessions on a 7-level scale. Results are stable along time.

Interestingly, the trends observed are similar to those in Amatista participants, with a greater emphasis on social presence over spatial presence and a decrease in perceived value when assessed using observable signs. However, unexpected observable signs are being analyzed, such as the disappearance of OCD during the session. Over time, there are no significant changes in responses, nor there is an apparent correlation between presence and either satisfaction or memory retention. However, the presence results are too close to the scale's edge to reliably correlate them with any other factors.

Qualitatively, users showed high level of enjoyment and satisfaction during the sessions. Comodia Elderly project will later continue with other immersive experiences including virtual visits to significant places and telepresence calls with people who are important to them. The overall effect in satisfaction and memory recall will be assessed in the final stage.

4 CONCLUSIONS

In Comodia Elderly project, we have shown that elderly with intellectual disability report high levels of spatial and social presence when exposed to immersive music content of events that they have previously attended live, even when some do not acknowledge having lived it. Reported presence is significantly stronger than those in elderly with neurodegenerative diseases. This suggests that virtual reality may be especially suited for this population, to broaden their emotional and experiential world, and obtain additional unexpected positive results that improve their quality of life.

ACKNOWLEDGMENTS

The authors wish to thank Miguel Garrido for this work in the video recording and editing. This work has been partially supported by the Spanish Administration agency Red.es (PRTR/NextGenerationEU) under project 2021/C005/00144164 COMODIA.

REFERENCES

- [1] G. De Graaf, F. Buckley, and B. G. Skotko. Estimation of the number of people with down syndrome in the united states. *Genetics in Medicine*, 19(4):439–447, 2017.
- [2] M. Lombard, T. B. Ditton, and L. Weinstein. Measuring presence: the temple presence inventory. In *Proceedings of the 12th annual international workshop on presence*, pp. 1–15, 2009.
- [3] M. Orduna, P. Perez, J. Gutierrez, and N. Garcia. Methodology to assess quality, presence, empathy, attitude, and attention in 360-degree videos for immersive communications. *IEEE Transactions on Affective Computing*, 14(3):2375–2388, jul-sep 2022.
- [4] P. Perez, E. Vallejo, M. Revuelta, M. V. Redondo Vega, E. Guervós Sánchez, and J. Ruiz. Immersive music therapy for elderly patients. In *ACM International Conference on Interactive Media Experiences*, pp. 47–52, 2022.